Table 5: Table sizes (number of rows and space in memory) in the seven considered benchmarks.

TPC-C	**Table**	**Total**	**Item**	**Stock**	**District**	**Warehouse**	**Customer**	**History**	**Order**	**NewOrder**	**OrderLine**
	Rows	599M	100k	120M	12k	1.2k	36M	36M	36M	10.8M	360M
	Size (MB)	355994	146	129110	5	0.49	61433	61433	6615	1987	95265
YCSB	**Table**	**Total**	**Account**								
	Rows	600M	600M								
	Size (MB)	0.564	0.564								
PPS	**Table**	**Total**	**Product**	**Part**	**Supplier**	**ProdPart**	**SupPart**				
	Rows	600M	25M	50M	25M	250M	250M				
	Size (MB)	59101	2950	6200	2950	23500	23500				
SmallBank	**Table**	**Total**	**Account**	**Checking**	**Savings**						
	Rows	600M	200M	200M	200M						
	Size (MB)	73296	35696	18800	18800						
MovR	**Table**	**Total**	**User**	**Vehicle**	**Ride**	**History**	**Code**	**UserCode**			
	Rows	480M	50M	25M	150M	150M	5M	100M			
	Size (MB)	434853	36072	18101	206835	58644	57600	57600			
DS Movie	**Table**	**Total**	**User**	**Movie**	**Review**						
	Rows	600M	600M	1k	0						
	Size (MB)	364801	364800	1	0						
DS Hotels	**Table**	**Total**	**User**	**Hotel**							
	Rows	525M	150M	375M							
	Size (MB)	87600	18600	69000							

A BENCHMARK SCHEMAS

Table 5 provides a per-table breakdown of the number of rows and memory volume occupied when we scale each of the seven existing benchmarks to approximately the same as TPC-C at 1200 warehouses.

B BENCHMARKING SCENARIOS

Figure 13, Figure 14, Figure 15, Figure 16, and Figure 17 present the complete profiling experiments across all evaluated systems (Calvin, SLOG, Detock, and Janus) discussed in Section 3.4. We broadly follow the methodology of Gaia [36]. Below, we detail the configuration of each evaluation scenario.

Scalability. The scalability experiment in Figure 13 executes default benchmark configurations while systematically increasing the input throughput until the throughput plateaus or p99 latency exceeds 1 second.

Access Patterns. To evaluate system sensitivity to geo-distribution, we sweep the fraction of FSH and MH transactions from 0% to 100% for operations that are not structurally constrained to LSH execution (Figure 14). Consequently, most static legacy benchmarks cannot reach 100% geo-distribution as their rigid schemas do not support geo-distribution across all transactions.

Network Instability. In Figure 15, we follow the default benchmark configurations while incrementally injecting additional latency and jitter between the two regions using NetEm. This emulates deployments spanning broader geographic distances than our baseline `us-west-1` and `eu-west-1` setup or operating over degraded network links.

Packet Loss. Analogous to the network instability scenario, we induce a 0–10% packet loss rate on the inter-region network link (Figure 16).

Sunflower Scenario. Finally, we execute the sunflower scenario on all four systems (Figure 17), gradually migrating the input load from one region to the other.

C MIMICKING EXISTING BENCHMARKS WITH PENTERACT

To reproduce the execution profiles of legacy benchmarks with Penteract, we adopt the parameter configurations detailed in Table 6. These parameter choices reflect the average key intensities identified earlier in our benchmark characterization, aggregating transactions belonging to the same transaction family. Additionally, we adjust key counts to account for schema payload discrepancies. If the legacy benchmark row sizes diverge significantly from Penteract's schema, we adjust the key footprint per transaction to compensate the critical path execution lengths. No temporal dynamics were used during this replication experiment.

D PENTERACT TRANSACTION FAMILY ABLATIONS

To complement the disaggregated latency analysis in Section 5.3, Figure 18 and Figure 19 provide the isolated throughput and latency breakdown across each individual transaction family.

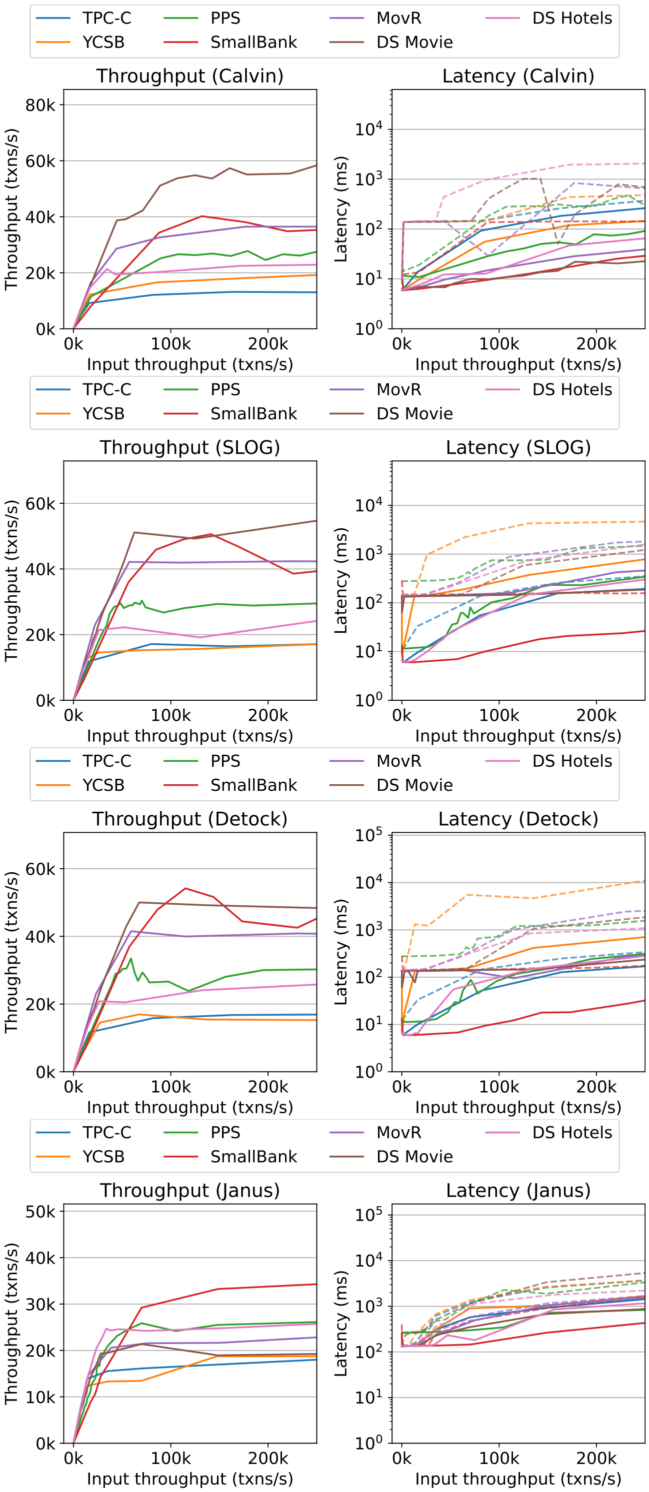

Figure 13: Throughput, p50 (solid lines), and p95 (dashed lines) latency on different benchmarks while varying the input throughput.

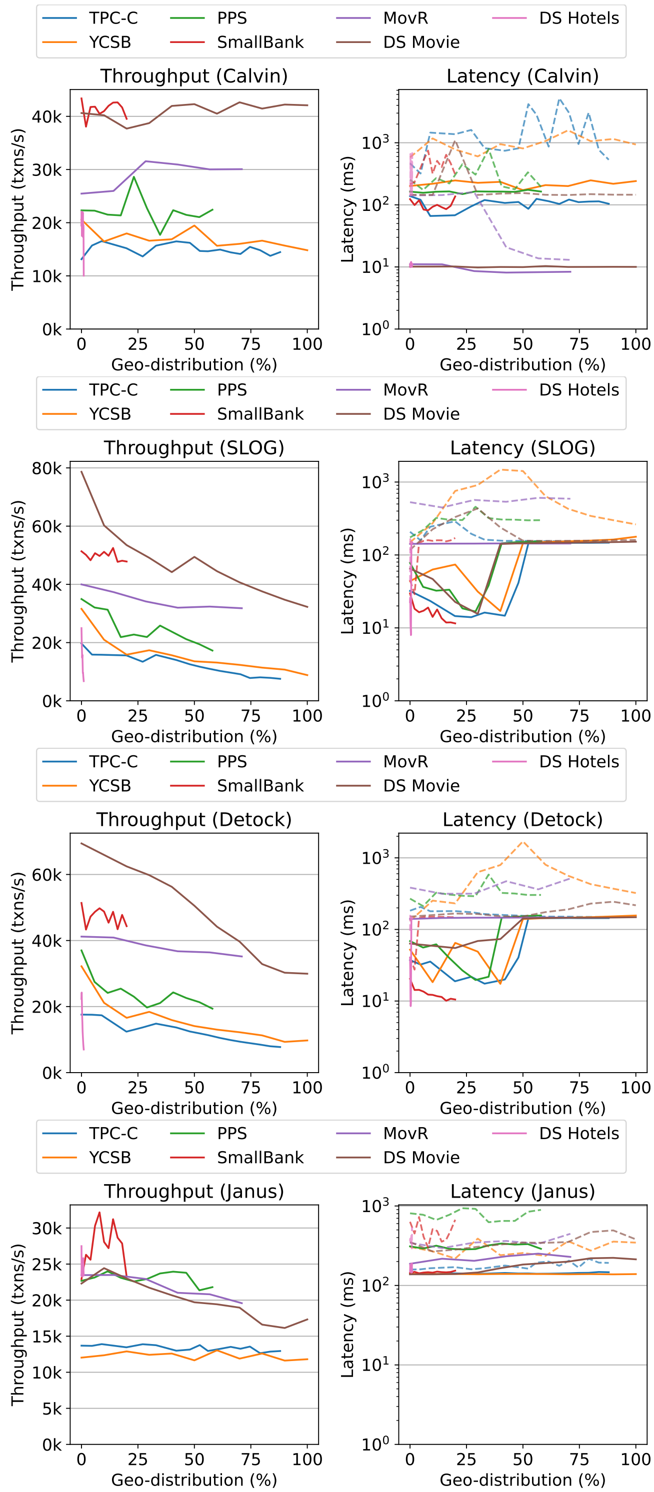

Figure 14: Throughput, p50 (solid lines), and p95 (dashed lines) latency on different benchmarks while varying the transaction geo-distribution.

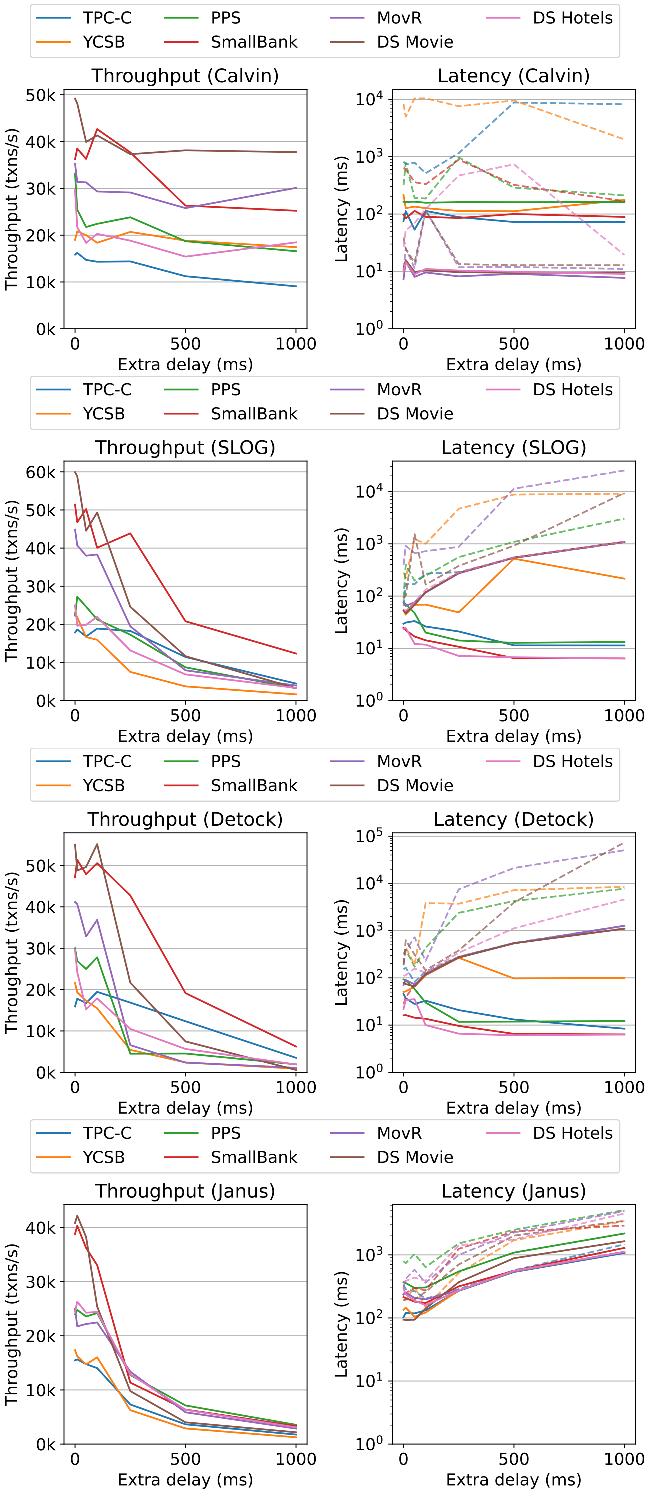

Figure 15: Throughput, p50 (solid lines), and p95 (dashed lines) latency on different benchmarks while adding additional cross-regional latency.

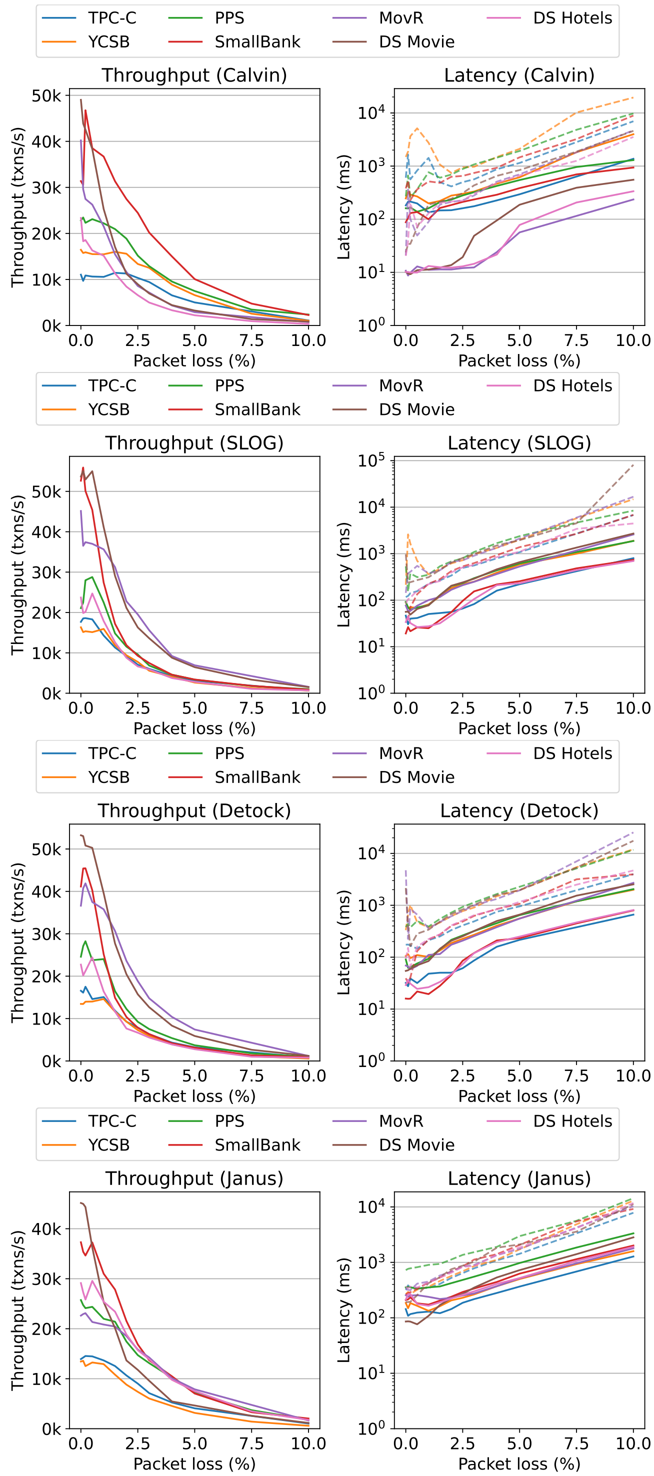

Figure 16: Throughput, p50 (solid lines), and p95 (dashed lines) latency on different benchmarks with increasing packet loss.

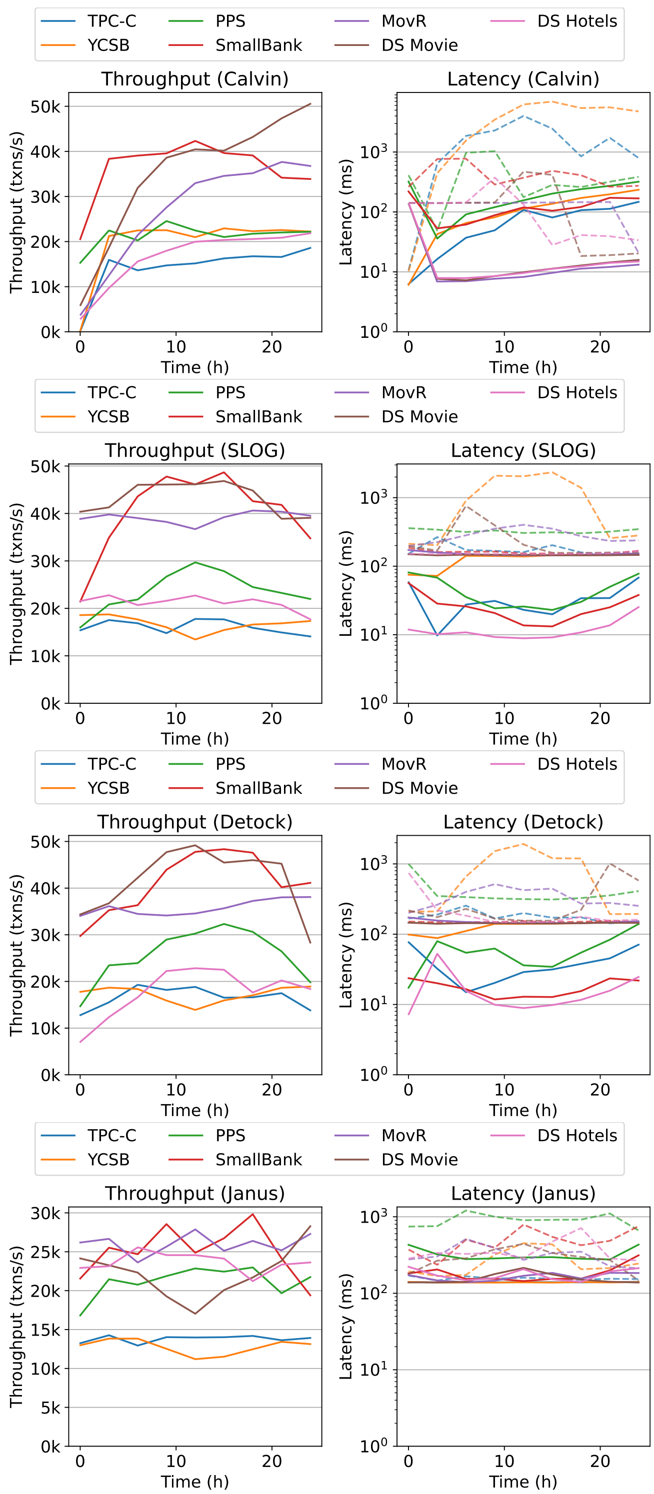

Figure 17: Throughput, p50 (full lines), and p95 (dashed lines) latency on different benchmarks in the sunflower scenario.

Table 6: Exact parameter vector mappings in Penteract to emulate legacy benchmarks. All 5-tuple vectors correspond to transaction families in order: (Short Read & Write : Short Read-Only : Large Read-Only : Insert-Only : Delete-Only). Foreign Single Home (FSH) and Multi-Home (MH) hit ratios are given per family (LSH = 100% - FSH - MH).

Emulated Benchmark	Transaction Mix (%) (SRW:SRO:LRO:INS:DEL)	Key Intensity (Keys / Txn)	FSH Ratio (%) (per family)	MH Ratio (%) (per family)	Dependency (%) (per family)	Contention
TPC-C [11]	92 : 4 : 4 : 0 : 0	7 : 1 : 1 : − : −	1 : 0 : 0 : − : −	4 : 0 : 0 : − : −	0 : 0 : 0 : − : −	0.0
YCSB [9]	100 : 0 : 0 : 0 : 0	10 : − : − : − : −	25 : − : − : − : −	25 : − : − : − : −	0 : − : − : − : −	0.0
PPS [51]	77 : 23 : 0 : 0 : 0	2 : 0.2 : − : − : −	0 : 0 : − : − : −	37 : 0 : − : − : −	75 : 0 : − : − : −	0.0
SmallBank [3]	80 : 20 : 0 : 0 : 0	1 : 0.1 : − : − : −	0 : 0 : − : − : −	13 : 0 : − : − : −	25 : 50 : − : − : −	0.0
MovR [27]	34 : 19 : 0 : 47 : 0	2 : 3 : − : 1 : −	0 : 0 : − : 6 : −	50 : 0 : − : 0 : −	0 : 0 : − : 0 : −	0.0
DS Movie [18]	100 : 0 : 0 : 0 : 0	1 : − : − : − : −	0 : − : − : − : −	50 : − : − : − : −	0 : − : − : − : −	1.0
DS Hotels [18]	1 : 99 : 0 : 0 : 0	7 : 4 : − : − : −	50 : 1 : − : − : −	0 : 0 : − : − : −	0 : 0 : − : − : −	0.0

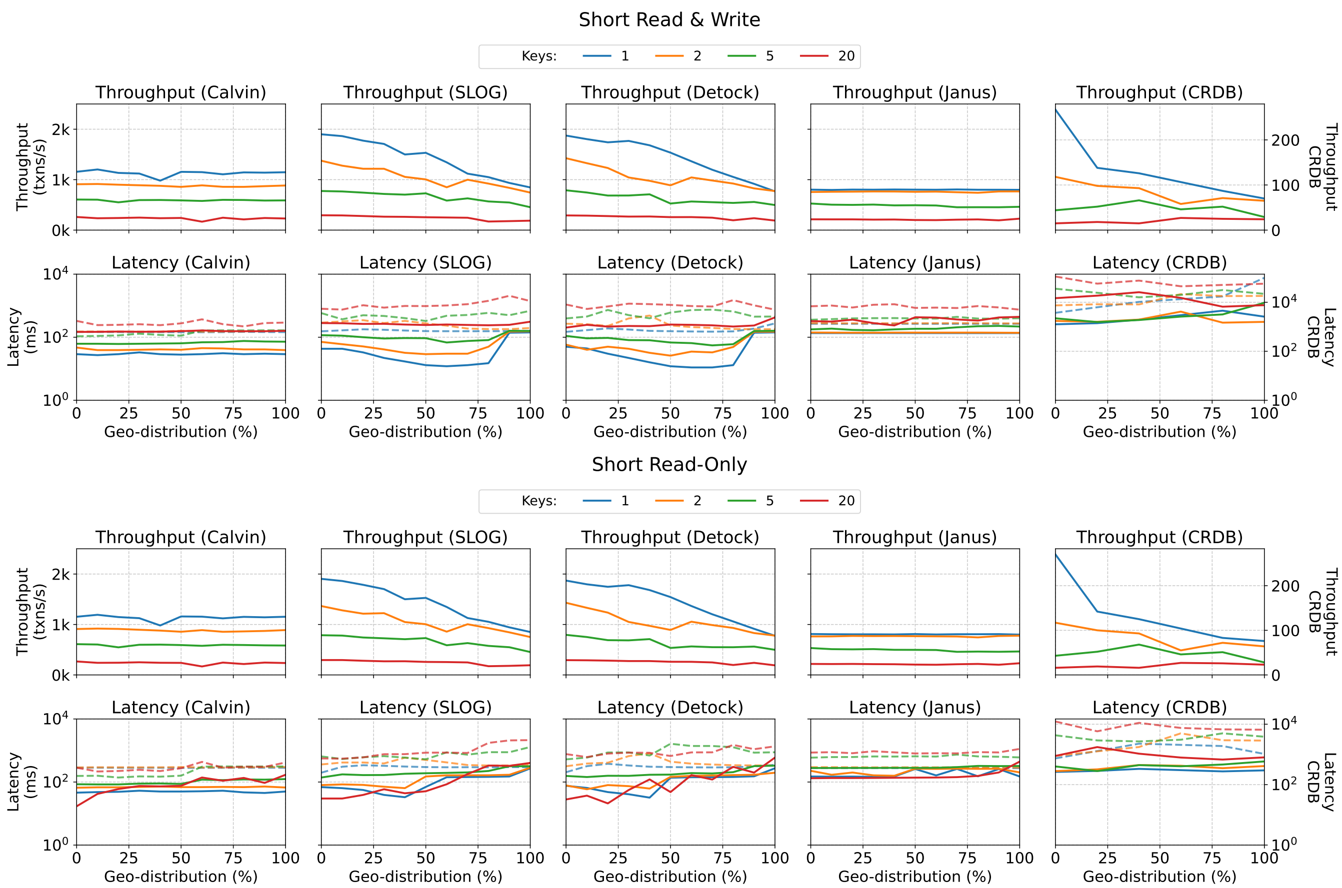

Figure 18: The Penteract throughput, p50 (solid lines), and p95 (dashed lines) latency for Short Read & Write and Short Read-Only transaction families.

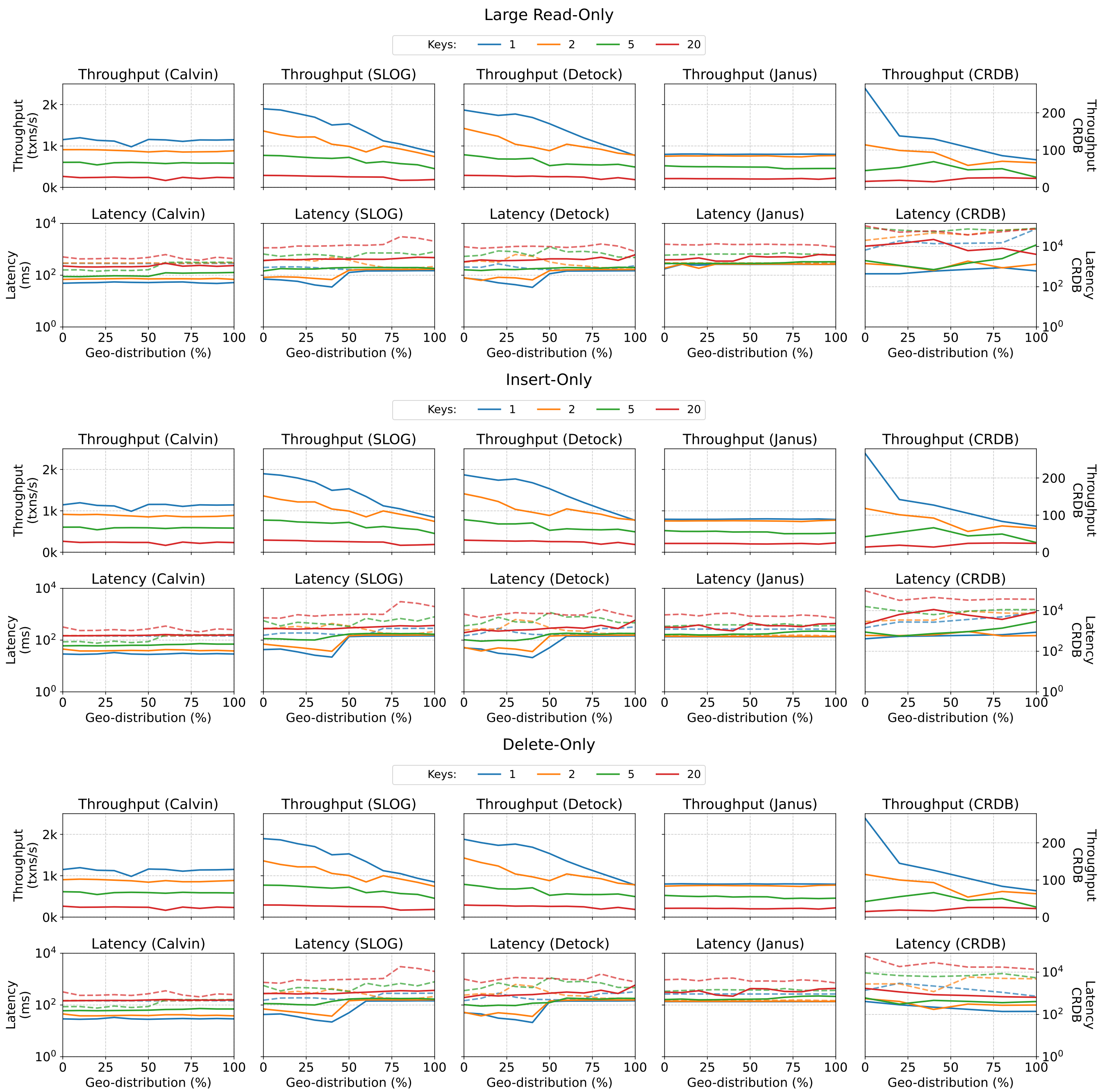

Figure 19: The Penteract throughput, p50 (solid lines), and p95 (dashed lines) latency for Large Read-Only, Insert-Only, and Delete-Only transaction families.